Chapter 11 Review

Campbell Biology, AP Edition, 12e

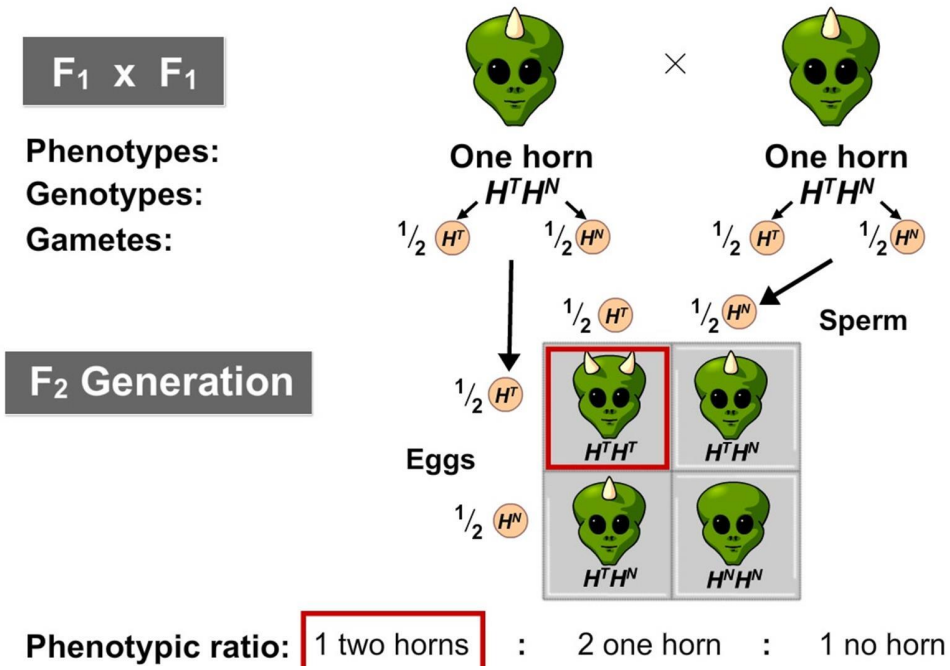
Chapter 11 Review

Summary of Key Concepts

Is all cell-to-cell communication regulated by the plasma membrane? What about apoptosis? **(BIG IDEA 3)**

Concept 11.1: External signals are converted to responses within the cell

- ********Signal transduction pathways** are crucial for many processes. Bacterial cells can sense the local density of bacterial cells (quorum sensing). Signaling during yeast cell mating has much in common with processes in multicellular organisms, suggesting an early evolutionary origin of signaling mechanisms.
- Local signaling by animal cells involves direct contact or the secretion of local regulators. For long-distance signaling, animal and plant cells use ****hormones**; animals also pass signals electrically.
- Like epinephrine, other hormones that bind to membrane receptors trigger a three-stage cell-signaling pathway:



Binding of a signaling molecule from the extracellular fluid to a receptor in the plasma membrane triggers reception, transduction, and response within the cell cytoplasm.

1. Signal reception. The signaling molecule in the extracellular fluid binds to a receptor in the plasma membrane.
2. Signal transduction. In this diagram, three relay molecules show a signal transduction pathway where each relay molecule brings about a change in the next molecule in the pathway.
3. Cellular response. The final molecule in the pathway in step 2 activates a cellular response.

What determines whether *a cell responds to a hormone such as epinephrine? What determines* how *a cell responds to such a hormone?*

Concept 11.2: Signal reception: A signaling molecule binds to a receptor, causing it to change shape

- The binding between signaling molecule (****__ligand__) and receptor is highly specific. A specific shape change in a receptor is often the initial transduction of the signal.
- Three of the major types of cell-surface transmembrane receptors are the following: (1) **G protein-coupled receptors (GPCRs)** work with cytoplasmic **G proteins**. Ligand binding activates the receptor, which then activates a specific G protein, which activates yet another protein, thus propagating the signal. (2) **Receptor tyrosine kinases (RTKs)** react to the binding of signaling molecules by forming dimers and then adding phosphate groups to tyrosines on the cytoplasmic part of the other monomer making up the dimer. Relay proteins in the cell can then be activated by binding to different phosphorylated tyrosines, allowing this receptor to trigger several pathways at once. (3) **Ligand-gated ion channels** open or close in response to binding by specific signaling molecules, regulating the flow of specific ions across the membrane.
- The activity of all three types of receptors is crucial; abnormal GPCRs and RTKs are associated with many human diseases.
- Intracellular receptors are cytoplasmic or nuclear proteins. Signaling molecules that are hydrophobic or small enough to cross the plasma membrane bind to these receptors inside the cell.

How are the structures of a GPCR and an RTK similar? How does initiation of signal transduction differ for these two types of receptors?

Concept 11.3: Signal transduction: Cascades of molecular interactions transmit signals from receptors to relay molecules in the cell

- At each step in a signal transduction pathway, the signal is transduced into a different form, which commonly involves a shape change in a protein. Many signal transduction pathways include ****__phosphorylation cascades__, in which a series of ****__protein kinases__ each add a phosphate group to the next one in line, activating it. Enzymes called ****__protein phosphatases__ remove the phosphate groups. The balance between phosphorylation and dephosphorylation regulates the activity of proteins involved in the sequential steps of a signal transduction pathway.
- ****__Second messengers__, such as the small molecule ****__cyclic AMP (cAMP)__ and the ion Ca2+, diffuse readily through the cytosol and thus help broadcast signals quickly. Many G proteins activate ****__adenylyl cyclase__, which makes cAMP from ATP. Cells use Ca2+ as a second messenger in GPCR and RTK pathways. The tyrosine kinase pathways can also involve two other second messengers, ****__diacylglycerol (DAG)__ and ****__inositol trisphosphate__IP3. IP3 can trigger a subsequent increase in Ca2+ level.

What is the difference between a protein kinase and a second messenger? Can both operate in the same signal transduction pathway?

Concept 11.4: Cellular response: Cell signaling leads to regulation of transcription or cytoplasmic activities

- Some pathways lead to a nuclear response: Specific genes are turned on or off by activated transcription factors. In others, the response involves cytoplasmic regulation.
- Cellular responses are not simply on or off; they are regulated at many steps. Each protein in a signaling pathway amplifies the signal by activating multiple copies of the next component; for long pathways, the total amplification may be over a millionfold. The combination of proteins in a cell confers specificity in the signals it detects and the responses it carries out. ****Scaffolding proteins** increase signaling efficiency. Pathway branching further helps the cell coordinate signals and responses. Signal response can be terminated quickly because ligand binding is reversible.

What mechanisms in the cell terminate its response to a signal and maintain its ability to respond to new signals?

Concept 11.5: Apoptosis requires integration of multiple cell-signaling pathways

- ****Apoptosis**is a type of programmed cell death in which cell components are disposed of in an orderly fashion. Studies of the soil worm *Caenorhabditis elegans* clarified molecular details of the relevant signaling pathways. A death signal leads to activation of caspases and nucleases, the main enzymes involved in apoptosis.
- Several apoptotic signaling pathways with related proteins exist in the cells of humans and other mammals, triggered in different ways. Signals eliciting apoptosis can originate from outside or inside the cell.

What is an explanation for the similarities between genes in yeasts, nematodes, and mammals that control apoptosis?